Cancer Health Disparities Among Indian and Pakistani Immigrants in the United States

A Surveillance, Epidemiology, and End Results-based Study From 1988 to 2003

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BACKGROUND. Immigrants from India and Pakistan comprise about 1.5% of the total United States population. To the author' knowledge, this is the first study analyzing epidemiologic data on Indian and Pakistani patients with newly diagnosed cancer.

METHODS. The Surveillance, Epidemiology, and End Results (SEER) database has reported cancer incidence from 1973 to 2003. Since 1988, the Indian and Pakistani population has been reported separately under race and ethnicity. Frequency and survival analyses with SEER data were performed, and descriptive parameters were calculated along with overall survival in common cancers.

RESULTS. In total, 6889 cases were reported between 1988 and 2003, and those cases included 51% men and 49% women (P = .821). Among men, 30% of cases were prostate cancers, 10% of cases were lung cancers, and 9% of cases were colorectal cancers. Among women, 38% of cases were breast cancers, 15% of cases were genital cancers, and 7% of cases were colorectal cancers. When overall survival with common cancer was compared between immigrants and non-Hispanic whites in the United States, it was better among Indian and Pakistani immigrants.

CONCLUSIONS. In a United States-based Indian and Pakistani population, prostate and breast cancers were the most common malignancies in men and women, respectively. This differed from the incidence in India and Pakistan, where oral cavity cancer in men and breast and cervical cancers in women were the most common. These immigrant cancer patients also had better survival. This change in demographics may be attributed to multiple factors, and the current data have implications on cancer screening and intervention. ***Cancer*** **2008;113:1423–30.**

KEYWORDS: Surveillance, Epidemiology, and End Results, Indian and Pakistani immigrants, cancer disparities, survival.

Health disparities are universal among different countries, sociocultural groups, races, and ethnicities. Studies that determine these disparities, such as migration studies, help to identify different groups with various diseases regarding their incidence, prevalence, diagnosis, treatments, and outcomes.[1,2] Some early migration studies were done among Japanese immigrants living in the United States in the post-World War II era.[3,4] Subsequent migration studies also were extended to Chinese and European immigrants and, later, to Vietnamese immigrants.[5,6] Among the demographics examined in those studies were differences in socioeconomic conditions, cultural

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background, and level of education.[7] In addition to those migration studies, anthropologic and ethnogenomic studies have given us a better idea about some of the disparities among immigrants.[8] On the basis of the findings from those health-disparity studies, therapeutic and preventive strategies can be tailored to different ethnic groups.

The Surveillance, Epidemiology, and End Results (SEER) Program is well recognized all over the world. It is considered the standard for methodological structure and quality among cancer registries around the world. Quality control and maintenance have been comprehensive, integral, and important parts of SEER since its inception. In this regard, each year, studies are conducted in SEER areas to evaluate and improve the quality and completeness of the data being reported.[9-14] The World Health Organization (WHO) also has its own statistical information system that collects data on all major cancers from almost all countries across the world. The incidence and prevalence vary from country to country based on age, sex, ethnicity, socioeconomic status, environmental conditions, and other demographics.[15]

The term 'Asian American' usually refers to individuals whose familial roots originate from many countries, ethnic groups, and cultures of the Asian continent, including but not limited to Asian Indian, Bangladeshi, Bhutanese, Burmese, Nepalese, Pakistani, Sikh, Sri Lankan, Thai, Cambodian, Chinese, Filipino, Indonesian, Japanese, Korean, Laotian, Malayan, Mien, Vietnamese, and many more.[9] According to U.S. Census data, there are slightly greater than 12 million Asians living in the United States. The Asian-American population consists of approximately 24% Chinese, 21% Filipino, 12% Japanese, 11.8% Asian Indian, 11.6% Korean, 8.9% Vietnamese, and 8% Pakistani.[9,15-19] Most Asian Americans who have arrived since 1965 reside mainly in 10 large metropolitan areas. In 1996, an estimated 4 in 10 Asian Americans lived in California.[17]

There have been numerous studies on the incidence of, prevalence of and survival from cancer among the Chinese, Filipino, and Japanese populations living in the United States.[1-6,20] The social, environmental, economic, and genetic influences on the incidence and prevalence of cancer among the populations described above have been studied well, whereas other Asian-American populations have not. In total, 1.3 billion individuals live in India and Pakistan.[15] The numbers of immigrants from those 2 countries are increasing in the United States; therefore, it is important to know the effect of demographics and migration on this population. The WHO, as discussed above, collects cancer incidence and prevalence data from different countries around the world, including India and Pakistan.[15] Several private and public organizations also collect data in these countries.

The Asian Indian and Pakistani populations comprise approximately 2.5 million individuals in the United States. Despite this large number of individuals, to date, no study has examined the incidence and prevalence of cancer or any descriptive parameters in this population. To our knowledge, this is the first study of its kind that examines these issues in this group.

MATERIALS AND METHODS

SEER started collecting data on cancer cases in January 1973.[21] The SEER registries routinely collect data on patient demographics, primary tumor site, morphology, stages at diagnosis, first course of treatment, and follow-up for vital status. The updated version of the SEER 'Cancer Incidence Public Use Database from 1973 to 2003' was released in April 2006. Since 1988, the incidence of cancer in Asian Indian and Pakistani population has been reported in a separate subcategory under race and ethnicity. Frequency and descriptive parameters were calculated carefully for analyses that included age, sex, marital status, SEER registry, vital statistics, stages, grades, tumor sites, and methods of confirmation.[21] Data were transferred to SAS software (version 10.0), and percentage were calculated. Student *t* tests were used to examine the differences. Because of the lack of a population denominator for these populations in Census data, incidence rates could not be calculated accurately. However, it is noteworthy that, based on the number of total Asian Indian and Pakistani populations reported from different nongovernment organizations (NGOs), the approximate incidence rates for different common cancers were calculated. Survival analyses on the available and relevant data were performed on the most common cancers in Asian Indians and Pakistanis and were compared with the non-Hispanic white cancer population reported in the same database. The Kaplan-Meier method was used for these analyses.[21] A 2-tailed P value $\leq .05$ was considered statistically significant.

RESULTS

Approximately 39% of the estimated Asian Indian and Pakistani population live in the 17 SEER registry areas. In total, 6889 cases were reported between 1988 and 2003 in Asian Indians and Pakistanis (approximately 0.03%). Of those cases, 3508 individuals (51%) were men, and 3381 (49%) were women

(P = .821). This distribution also was comparable (P = .623) with cases in the non-Hispanic white population reported in the SEER database from 1988 to 2003. The average age at diagnosis was 52 years (51 years for men, 53 years for women; P = .7295). Although the average age at diagnosis for non-Hispanic whites reported in the SEER database was 56 years, the difference was not statistically significant (P = .116). Eighteen percent of cases were reported from the San Francisco Registry, and 16% were reported from the New Jersey Registry. Approximately 56% of the cases reported were from all registries in California. It is estimated that approximately 36% of all Asian populations and 35% of the Asian Indian and Pakistani population reside in this area.

When SEER started reporting Asian Indians and Pakistanis in a separate subcategory in 1988, only 109 cases (2%) were reported. However, by 2003, there were 1106 cases (16%). It is noteworthy that, from 1988 to 1999, in total, 2763 cases (40%) were reported. After including more areas in greater California and including the New Jersey registry, the total number of reported cases increased significantly. From 2000 to 2003, in total, 4126 cases (60%) were reported in this subcategory of race and ethnicity. Approximately 26% of the cases were reported only in 2002 and 2003. In addition, 96% of the cases were reported from hospital sources.

Table 1 shows that, among men, prostate cancer (1062 cases; 30%) was the most common cancer followed by lung cancer (337 cases; 10%) and colorectal cancer (324 cases; 9%). Among women, breast cancer was the most common (1283 cases; 38%) followed by genital cancer (506 cases; 15%) and colorectal cancer (222 cases; 7%) (Tables 1, 2). The estimated incidence rates based on the population denominator from NGOs are 152 cases of prostate cancer, 56 cases of lung cancer, and 52 cases of colorectal cancer per 100,000 population; and the estimated incidence

TABLE 1
Asian Indian and Pakistani Cancer Cases Reported in the Surveillance, Epidemiology, and End Results Database From 1988 to 2003 According to Anatomic Site*

	No. of Cases (%)		
Anatomic Site	All Cases	Men	Women
Total	6889 (100)	3508 (100)	3381 (100)
Breast	1296 (19)	13 (0.5)	1283 (38)
Prostate	1062 (15)	1062 (30)	0 (0)
Colon/rectum	546 (8)	324 (9)	222 (7)
Lung/bronchus	490 (7)	337 (10)	153 (5)
Lymphoma	460 (7)	278 (8)	182 (5)
Female genital organ	506 (7)	0 (0)	506 (15)
Leukemia	318 (5)	193 (6)	125 (4)
Urinary system	355 (5)	256 (7)	99 (3)
Endocrine	258 (4)	74 (2)	184 (5)
CNS	200 (3)	115 (3)	85 (3)
Oropharynx	230 (3)	159 (5)	71 (2)
Stomach	164 (2)	107 (3)	57 (2)
Liver	129 (2)	89 (3)	40 (1)
Pancreas	117 (2)	71 (2)	46 (1)

CNS indicates central nervous system.
*From the Surveillance, Epidemiology, and End Results Program of the National Cancer Institute public use database from 1973 to 2003 (released in April 2006, but analyses were performed with the data from 1988-2003).

TABLE 2
The Incidence of Different Cancers Complied From Databases of the National Cancer Institute (NCI), the NCI Surveillance, Epidemiology, and End Results Program, and the World Health Organization*

Non-Hispanic White		Asian Indian/ Pakistani (in USA)		Indian Population (in India)		Pakistani Population (in Pakistan)	
Cancer Type	%	Cancer Type	%	Cancer Type	%	Cancer Type	%
Men		Men		Men		Men	
Prostate	33	Prostate	30	Oral cavity	22	Oral cavity	17
Lung/bronchus	13	Lung/bronchus	10	Lung	9	Lung	14
Colon/rectum	10	Colon/rectum	9	Esophagus	7	Larynx	6
Women		Women		Women		Women	
Breast	32	Breast	38	Cervical	30	Breast	34
Lung/bronchus	12	Female genital	15	Breast	19	Oral cavity	10
Colon/rectum	11	Colon/rectum	7	Oral cavity	7	Ovary	7

*The NCI (Bethesda, Md); the NCI Surveillance, Epidemiology, and End Results Program public use database from 1973 to 2003 (released in April 2006, but analyses were performed with data from 1988-2003); and GLOBOCAN 2000 data on cancer incidence, mortality, and prevalence worldwide from the World Health Organization.

TABLE 3
The Incidence of Major Cancers in the Asian Indian and Pakistani Population Reported in the Surveillance, Epidemiology, and End Results Database From 1988 to 2003*

	All Cases		Lung Cancer		Colon/Rectal Cancer		Breast Cancer		Prostate Cancer	
	No.	%	No.	%	No.	%	No.	%	No.	%
Year										
1988-1993	819	12	41	8	68	12	144	11	104	10
1994-1998	1545	22	102	21	120	22	310	24	252	24
1999-2003	4525	66	347	71	358	66	842	65	706	66
Total	6889	100	490	100	546	100	1296	100	1062	100

*From the Surveillance, Epidemiology, and End Results (SEER) Program of the National Cancer Institute public use database from 1973 to 2003 (released in April 2006, but analyses were performed with the data from 1988-2003).

rates for women are 122 cases of breast cancer, 55 cases of genital cancer, and 32 cases of colorectal cancer per 100,000 population. Table 3 shows progressive increases of incidences of cancers in these minority groups.

Overall survival also was calculated by using survival analyses for 4 major cancers reported in Asian Indians and Pakistanis in the SEER database using the Kaplan-Meier method (Table 4). Breast cancer had the highest 1-year survival rate of 98.1%. This 1-year survival rate for breast cancer was comparable between Asian Indians and Pakistanis and non-Hispanic whites (98.1% vs 94.9%); and, even after 5 years, the survival rate remained comparable (80.3% vs 74.9%). The 1-year survival rate for prostate cancer was comparable between Asian Indians/Pakistanis (96.7%) and non-Hispanic whites (93.9%). However, the survival rate at 5 years was significantly higher in Asian Indians and Pakistanis than in non-Hispanic whites (81.8% vs 70.2%; $P < .001$). The survival rate for colorectal cancer was significantly higher both at 1 year (85.3% vs 76.6%; $P < .001$) and at 5 years (65.7% vs 46.9%; $P < .0001$). Lung cancer, as expected, had the lowest 1-year and 5-year survival rate among the other commonly reported cancers; however, the survival rate at both 1 year and 5 years was better among Asian Indians and Pakistanis (54.5% vs 42.1% $P < .001$) compared with non-Hispanic whites (22.7% vs 15.4%; $P < .01$).

DISCUSSION

Asian Indians and Pakistanis comprise a significant proportion (20%) of all Asians living in the United States.[15] Among the men who are immigrants from India and Pakistan, the most common cancer is prostate (30%). This also is true for non-Hispanic white Americans (33%). However, cancer of the oral cavity is the most common cancer in men both in India (22%) and in Pakistan (17%).[15] This could be because of the risk factors that are shared by these populations (Asian Indian/Pakistani immigrants and non-Hispanic whites), including mainly social and environmental factors.[22-27] This finding also correlates with the conclusions of other studies that the incidence of prostate cancer is high in Northern Europe and North America, intermediate in South America and in the rest of the Europe, and low in Asia and that immigrants often have the same risk as individuals who are natives in the country to which they migrate.[27-31]

Lung cancer is the second most common cancer among men in the Asian Indian and Pakistani population reported in the SEER database (10%). It is the second most common cancer in both India (9%) and Pakistan (14%) and in non-Hispanic whites in the United States (13%). This lower incidence in the immigrant populations can be attributed to the differences in smoking habits and environmental exposures among white Americans and immigrants from the Indian subcontinent.[7,10-14,32]

In India, cancer of the esophagus (7%) is the third most common cancer; and, in Pakistan, cancer

TABLE 4
Overall Survival in the Major Cancers Among Non-Hispanic Whites and Asian Indians and Pakistanis Reported in the Surveillance, Epidemiology, and End Results Database From 1988 to 2003*

	Survival Rate (SE), %			
Survival†	Colon/Rectal Cancer	Lung Cancer	Breast Cancer	Prostate Cancer
1 Year				
White	76.6 (0.1)	42.1 (0.1)	94.9 (0)	93.9 (0)
AIP	85.3 (1.7)	54.5 (2.5)	98.1 (0.4)	96.7 (0.6)
2 Years				
White	65.2 (0.1)	26.3 (0.1)	89.6 (0)	87.8 (0.1)
AIP	78.2 (2.1)	37.4 (2.6)	94.1 (0.8)	93.5 (0.9)
3 Years				
White	57.3 (0.1)	20.5 (0.1)	84.2 (0.1)	81.6 (0.1)
AIP	72.0 (2.4)	29.0 (2.6)	89.0 (1.1)	90.7 (1.1)
4 Years				
White	51.5 (0.1)	17.4 (0.1)	79.4 (0.1)	75.8 (0.1)
AIP	68.7 (2.6)	25.5 (2.6)	85.2 (1.3)	87.2 (1.4)
5-Years				
White	46.9 (0.1)	15.4 (0.1)	74.9 (0.1)	70.2 (0.1)
AIP	65.7 (2.8)	22.7 (2.6)	80.3 (1.6)	81.8 (1.8)

SE indicates standard error; AIP, Asian Indians and Pakistanis.
*From the Surveillance, Epidemiology, and End Results (SEER) Program of the National Cancer Institute public use database from 1973 to 2003 (released in April 2006, but analyses were performed with the data from 1988-2003).
†Survival in the non-Hispanic white cancer population and in the AIP cancer population as reported in the SEER database from 1988 to 2003.

of the larynx (6%) is the third most common cancer. However, colon cancer is the third most common cancer both in the non-Hispanic white population (10%) and in the Asian Indian and Pakistani populations (9%) in the United States. This is probably because of their similar risk factors from living in the same country. There also are documented differences in environment, culture, diet, health system access, healthcare maintenance, and colorectal cancer screening recommendations between the Indian subcontinental population and the United States population.[14,33-37] These differences in incidences also may be attributed to betel quid chewing, bidi smoking, high nonrectified alcohol consumption, and low socioeconomic conditions in India and Pakistan.[38-42] Studies have demonstrated that these behaviors are much less prevalent among Asian Indian and Pakistani immigrants to the United States, who also have higher socioeconomic status than the populations in Pakistan and India, other Asian immigrants, and all other minorities.[43-45]

Breast cancer is the most common cancer among women in both non-Hispanic white Americans (32%) and immigrants from India and Pakistan (38%). In Pakistan, although breast cancer (34%) is the most common type of cancer among women, in India, it is the second most common (19%). This is probably because of differences in early menarche, late menopause, nulliparity, first live birth in late ages, diet, lifestyle, and lactation practices.[5,16,19] Cervical cancer (30%) is the most common cancer type among women in India; however, once they migrate to the United States, breast cancer becomes the most common cancer. This may be explained by better cervical cancer screening programs performed in the United States and the sexual practices and level of education within the immigrant group. Other reasons may include the wide prevalence of human papillomavirus (HPV) in India and the relation between socioeconomic conditions and cervical caner.[28,29,46-48] Lung cancer is the second most common cancer among non-Hispanic white women in the United States; however, in the Asian Indian and Pakistani population, among women, cancers of the genital organs are the second most common. This can be explained by the differences in screening, smoking habits, and sexual practices. Colon cancer is the third most common cancer type in both the non-Hispanic white population and these immigrant populations.[46-48]

Overall survival also was better among the Indian and Pakistani immigrant subgroup than among non-Hispanic whites reported in the SEER database for common cancers (eg, colon, lung, breast, prostate). Several factors can be attributed to these differences, including but not limited to possible earlier diagnosis, more favorable tumor grade or stage at the time of diagnosis, healthier lifestyle, higher socioeconomic condition etc. It has been reported that this subgroup has easy access to healthcare because of the geographic locations where they live and also because they are among the groups with a higher percentage of insurance coverage.[5,9,16,18]

Among the chronic diseases, only cancer is unique, in that statistical data have been collected from populations uniformly and repeatedly across the 5 continents from 214 population groups living in 60 countries.[3,8] The SEER Program of the National Cancer Institute is a very well known authoritative source of information on cancer incidence and survival in the United States.[16,21] It is noteworthy that all of the SEER registries cover approximately 26% of the United States population. Regarding race and ethnicity, this includes 53% of Asians, 23% of African Americans, 70% of Hawaiian/Pacific Islanders, 42% of American Indians and Alaska Natives, and 40% of Hispanics.[9,16-19,21] It is also the only source of population-based information, both historic and current, on patient survival and disease stage.[9,17,21]

Because the SEER Program is considered the standard for methodological structure and quality among cancer registries all over the world, using this database provides reliability and accountability to the study. There is also a well documented difference between the immigrant populations and the local populations in India and Pakistan in terms of socioeconomic status, environmental exposures, different health behaviors, education, access to screening, early detection, and treatments.[13,14,33] Between 1975 and 2003, numerous studies were published using the SEER database. These studies compared patterns of cancer incidence among non-Hispanic whites, different minorities groups, immigrant groups, and matched controls. The studies used data from SEER, from private and regional cancer registries in the United States, from cancer registries in other countries (both government and private), and from the WHO. The conclusions from those studies were remarkably uniform.[13,14,33-35]

From the census data demographic information, immigration patterns, and SEER database demographic information, it is evident that the cases of reported cancers in the Asian Indian and Pakistani population are among immigrants.[1,2,49] Almost all of those studies concluded that cancer incidence patterns among first-generation immigrants were identical or nearly identical to those of their native countries. However, from the second-generation onward, these patterns changed and slowly evolved to resemble the cancer incidence patterns observed in the United States population. This was especially true for cancers related to hormones, such as ovarian cancer, breast cancer, prostate cancer, neoplasms of the uterine corpus, and cancers that may be attributable to westernized diets, such as colorectal malignancies.[35-39] The longer Asian individuals lived in the United States, the lower their rates of cancers that could be attributed to Asian diets, such as stomach cancer associated with the highly salted and nitrite-containing foods common in Asia; stomach cancer caused by *Helicobacter pylori*; cancers caused by infections, such as liver cancer caused by hepatitis B and C; cervical cancer caused by HPV; and cancers caused by specific environmental factors, such as nasopharyngeal cancer associated with significant and continued exposure to smoke from stoves used for cooking at home and salivary cancer related to cold, dark environments that produce vitamin A deficiencies.[40-42] The populations studied included first- and second-generation Vietnamese-Americans; Japanese immigrants living in Hawaii; Asian-American women; Korean Americans; Hmong refugees from Vietnam, Laos, and Thailand who settled in California after the Vietnam War; Pacific Islanders; and Alaska Natives.[4,7,30,31] With the help of those studies, medical and scientific communities were able to identify possible environmental factors related to cancer development and, thus, paved the way for the field of early cancer diagnosis and prevention. If migration studies can be performed on the second-generation Asian Indian and Pakistanis population in the future, then information can be complied about the effect of migration on these populations on cancer incidence, prevalence, and survival. On the basis of the other migration studies and the current age structures for second-generation Asian Indians and Pakistanis, it is estimated that these migration studies could be performed in the next 15 to 20 years.[5,46,48]

To our knowledge, this is the first study of its kind examining demographics and cancer incidence in the Asian Indian and Pakistani population residing in the United States. The survival data also add more information to our overall knowledge of cancer survival in minorities; however, this report includes the incidence but not the prevalence. Data from the Census Bureau are used as denominators to calculate the rate, although no complete data are available on the Asian Indian and Pakistani population to date.[17] The lack of available population denominator data preclude the calculation of incidence rates. Without data on the age structure of the United States Asian Indian and Pakistani population, it is not possible to separate the effects of differences in age versus risk factor prevalence on incidence. But it should be emphasized that, although this is a limitation, we have analyzed the data in the only possible way, and the results still are a valuable contribution to the epidemiologic literature and can have practical use in prioritizing cancer control opportunities for this population.

CONCLUSION

Prostate and breast cancers are the most common in men and women in the Indian and Pakistani immigrant population in the United States. It is important to note that immigrants from India and Pakistan who live in the United States have changes in their demographics regarding the incidence of different cancers and follow the trend in the United States population. The different factors that may be responsible for this migration effect include but are not limited to sociocultural, financial, educational, and environmental factors. A reduction in the overall cancer mortality and morbidity rates among minority populations, especially the minorities that are under studied and have less access to healthcare, would have a substantial impact on known cancer statistics and inferences. Different cultural barriers to cancer diagnosis and treatment and to preventive healthcare advice may take on positive added dimensions as the information and techniques of genetic and molecular epidemiology are applied increasingly to identify minority individuals and family members at high risk for cancer. Currently, physicians increasingly are involved and are called upon to deal with culturally diversified, sensitive issues, such as possible prophylactic surgery for family members and/or major changes in lifestyle and beliefs, possibly even including childbearing. Therefore, it would be most practical and imperative to have proper epidemiologic knowledge in these areas. This understanding can be achieved best by appropriate research and career development with existing and future knowledge. However, we need further studies to examine the epidemiology of cancers among the immigrant population and to identify the factors that are responsible for these effects. The results from such research will have implications for education, screening, and preventive strategies in the Asian Indian and Pakistani immigrant population.

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